

SkillCraft Technology – Data Science Internship Report

Task 01: Population Distribution Visualization

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Submitted To

SkillCraft Technology – Data Science Internship Program

1. Introduction

The objective of this task is to analyse country-wise population data and visualize the distribution using Python. The study focuses on identifying the most populous countries and understanding how population is distributed globally.

The project uses data visualization techniques to represent insights in a simple and interpretable way.

2. Dataset Description

The dataset used is geo_population.csv, which contains country-wise population data.

Key Features:

- **Country:** Name of the country
- **Population:** Total population of the country

The dataset helps in comparing population sizes across different countries.

3. Tools and Technologies Used

- Python (Programming language)
- Pandas (Data manipulation)
- Matplotlib (Data visualization)
- Visual Studio Code (Development environment)
- Git & GitHub (Version control)

4. Methodology

Step 1: Data Loading

The dataset was loaded using Pandas for analysis.

Step 2: Data Processing

- Sorted countries based on population
- Extracted top 10 most populous countries
- Prepared data for visualization

Step 3: Data Visualization

Two visualizations were created:

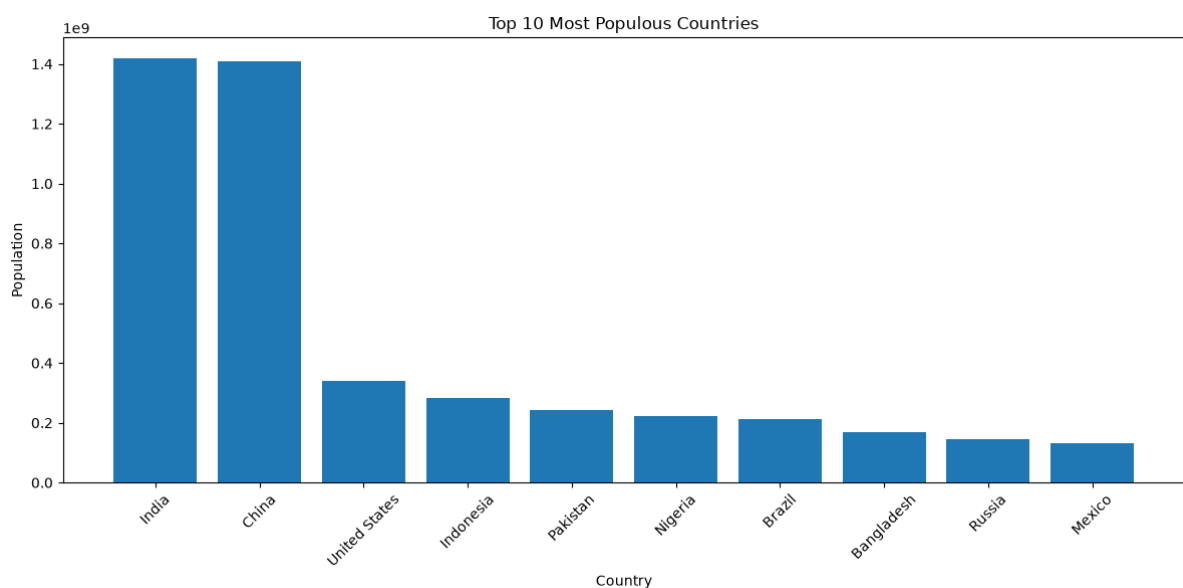
- Bar chart for top 10 countries
- Histogram for overall population distribution

5. Visualizations and Results

5.1 Bar Chart – Top 10 Most Populous Countries

The bar chart compares the population of the top 10 countries.

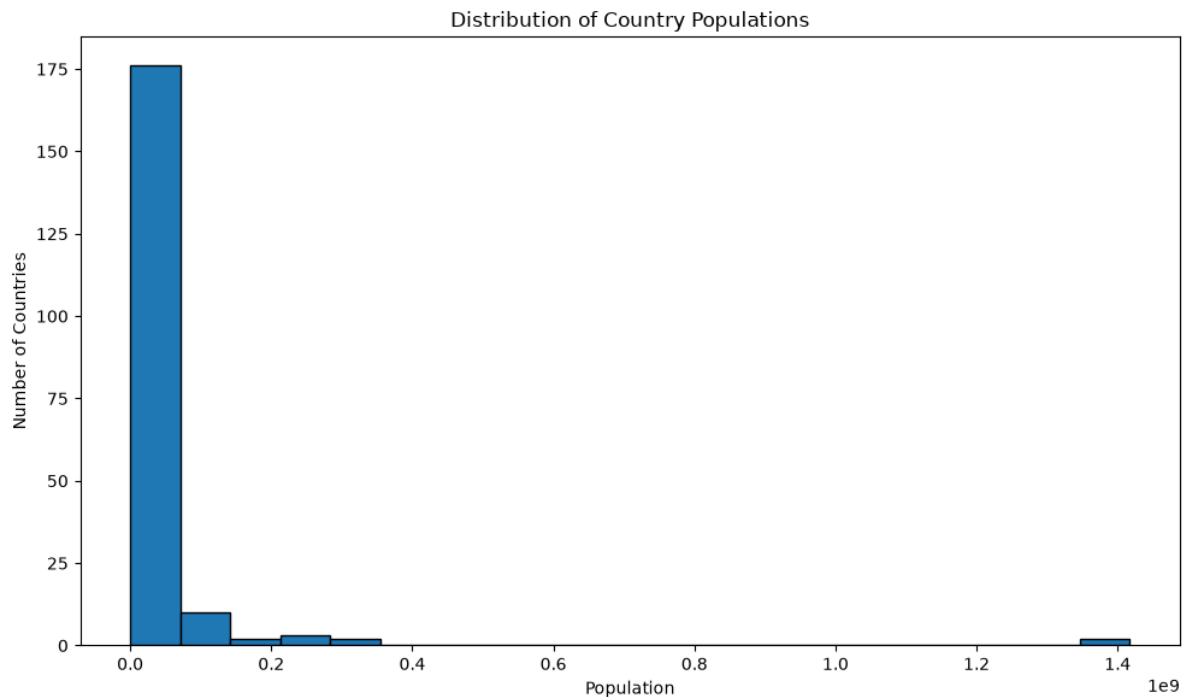
👉 India and China dominate the global population distribution, significantly higher than other countries.



5.2 Histogram – Population Distribution

The histogram shows how population values are distributed across all countries.

👉 The distribution is right-skewed, meaning most countries have smaller populations while a few countries have very large populations.



6. Key Insights

- India and China are the most populous countries in the dataset.
- A small number of countries account for a large portion of the global population.
- Most countries have relatively low population compared to top countries.
- Population distribution is highly uneven (right-skewed).

7. Conclusion

This project helped in understanding how population is distributed across countries using data visualization techniques. Bar charts and histograms effectively highlighted inequality in population distribution. The exercise improved skills in data analysis, visualization, and interpretation using Python.

8. Learning Outcomes

- Learned how to analyse CSV datasets using Pandas
- Created meaningful visualizations using Matplotlib
- Understood population distribution patterns
- Improved ability to interpret data visually
- Gained experience in structuring a data science report